01 / VERIFICATION PROBLEM

A mind map is useful only when its claims can be checked

Long PDFs make it difficult to remember where a conclusion came from. A clean diagram can hide omissions or incorrect grouping if the reader cannot return to the supporting page.

The verification goal is simple: preserve enough source location data that a reader can compare each important map node with the original document.

CONTROLLED CHECKS

01	Coverage
02	Hierarchy
03	Source page
04	Editable result

Expected map branch

A mind map is useful only when its claims can be checked - source badge p.1

02 / PAGE-AWARE METHOD

Extract text by page before building the hierarchy

The document is read page by page. Paragraphs inherit the page number they were extracted from, and chunks keep that location before any AI generation begins.

The model refers to stable chunk identifiers. The application resolves those identifiers back to page numbers, so the model does not invent page citations itself.

CONTROLLED CHECKS

01	Page 1 -> chunk A
02	Page 2 -> chunk B
03	Chunk ID -> source page
04	Node -> page badge

Expected map branch

Extract text by page before building the hierarchy - source badge p.2

03 / EVALUATION

Test completeness, placement and provenance separately

Completeness asks whether every major section appears. Placement asks whether details sit beneath the correct parent. Provenance asks whether a page badge points to text that actually supports the node.

A map can pass one test and fail another. Keeping these checks separate makes the result easier to diagnose and improve.

CONTROLLED CHECKS

01	Major sections present
02	Branches correctly nested
03	Page badges verified
04	Failures recorded

Expected map branch

Test completeness, placement and provenance separately - source badge p.3

04 / DECISION LOG

Keep, revise or reject the result using recorded evidence

After verification, record the sample version, generated output, test date and observed failures. A result should be kept only when its hierarchy is useful and its important claims can be traced back.

If the map omits a section or points to the wrong page, revise the method and run the same fixed sample again. A stable input makes changes comparable over time.

CONTROLLED CHECKS

01	Keep
02	Revise
03	Reject
04	Run again

Expected map branch

Keep, revise or reject the result using recorded evidence - source badge p.4